Unit7 How the macroeconomy works

7.1 The circular flow of income

- What national income measures
  - National income: the monetary value of all the goods and services that are produced by an economy in a given period of time
  - output method
    - The total value of output, or national income, is calculated by summing the value added by each industry to total output.
      - value added: the difference between the value of an industry's output and the and the value of inputs it buys from other firms
    - avoid double counting
  - income method
    - The monetary value of output must create an equal amount of income
    - calculated by adding up all the incomes earned by factor of production
  - expenditure method
    - National income=AD=C+I+G+X-M
    - output=income=expenditure
  - difference between GDP GNP NNP
    - GNP = GDP + net income from abroad
    - NNP = GNP – depreciation
    - National income at factor cost = national income at market prices – indirect taxes + subsidies
    - NNP at factor cost=NNI
- difference between nominal income and real national income
  - If national income is measured in real terms, the effects of inflation, or deflation, have been removed
  - Changes in real national income show changes in the total output of the economy
- Real national income as an indicator of economic performance
  - increase in real national income — more goods and services have been produced and therefore more needs and wants can be satisfied — improvement in the performance of the economy — if the population has increased, average living standards may not have risen
- circular flow of income model
  - shows the interconnections between different economic agents — illustrates how money and goods/services circulate within an economy — highlight the flow of income and expenditure
  - national output ≡ national income ≡ national expenditure
  - Introducing injections and withdrawals
    - withdrawls: part of the household income that is not spent on goods and services produced by the economy
    - injection: expenditure that is added to and increases the circular flow of income in an economy

A simple two-sector model of a closed economy, with saving and investment

  - Equilibrium income in a two-sector economy
    - C+I=Y — everything that has been produced will be sold
    - C+I>Y — S > I — unsold goods — cut back on production — reduces national income
    - C+I<Y — I > S — produce more to meet increasing demand — increased national income
    - equilibrium income
      - two-sector economy — saving=investment
      - open economy with the government
        - aggregate demand=national output
        - injections=withdrawls
  - Equilibrium income in an economy with government and foreign trade sectors

A model of an open economy with a government sector

    - AD = C + I + G + X – M — AD = C + I + G + （X – M）
  - Full employment income
    - the level of national income/output that creates enough jobs for everyone who is willing and able to work at current market wage rates
    - equilibrium level of national income<full employment level of national income — involuntary unemployment
  - effects of changes in injections and withdrawals on national income
    - increase in injections into circular flow of income — increase in aggregate demand — firms will increase their output and unemployment will fall
    - increase in withdrawls — decrease in aggregate demand — reduce the equilibrium level of national income
    - Higher taxes and more spending on imports — reduce the consumption of home-produced goods and services

7.2 Aggregate demand and aggreagate supply analysis

- changes in the price level are represented by movements along the AD and AS curves
  - movements along the AD curve
    - aggregate demand curve: shows the relationship between the price level and total spending in an economy
    - why a fall in the price level is likely to increase aggregate demand
      - home-produced goods and services will become relatively cheaper — imports sold fall and exports rise — increasing aggregate demand
      - increase the real value of asset that have a fixed monetary value — increase wealth — increase aggregate demand
  - movements along the AS curve
    - rise in the price level — more profitable to produce — incentives to increase output
    - AS curve is horizontal — a lot of spare capacity — increase in aggregate demand will ecourage firms to produce more — even if price level remains same
    - AS curve is vertical — full capacity — firms are unable to produce more outout in response to the increase in aggregate demand
      - short-run — national income cannot increase beyond this level ofmoutput
      - long-run — productive capacity may increase
- factors that shift the AD curve and the short-run aggregate supply(SRAS) curve
  - shifts in the AD curve
    - Changes in real incomes and employment
    - Changes in government spending, taxation and borrowing — government borrows to finance spending — stimulate economic activity — rightward shift in AD curve
    - Changes in monetary policy interest rates
    - Changes in the external value of a country's exchange rate
      - depreciation of the exchange rate — exports cheaper and imports more expensive — increasing net exports — shift AD curve to the right
      - appreciation has the opposite effect
    - Changes in the rate of economic growth of trading partner nations — trading partners experience economic growth — demand more exports
    - Fluctuations in consumer and business confidence — optimistic about future economic conditions — increase investment — rightward shift in AD curve
  - shift in the SRAS curve
    - Labour productivity
    - Key raw material and component prices
    - Business indirect taxes, subsidies, VAT, environmental & employment taxes
    - Cost of imported materials — affected by movements in a country's exchange rate + fluctuations in global prices
    - Changes in consumer and producer confidence
    - Higher wage costs perhaps arising from a higher minimum wage
- factors that affect long-run aggregate supply
  - normal capacity level of output: the maximum output that an economy can continue to produce in the long run without experiencing any inflationary pressure — short-run — able to produce more than its normal capacity level of output — long-run — shortages of labour and other factors of production — excess demand — rising inlation
- why underlying economic growth is representated by a rightward shift in the LRAS curve
  - increase in the normal capacity level of output of the economy — rightward shifts in LRAS curve
  - increase in the productive capacity of the economy can be caused by two factors:
    - increase in the supply of the factors of production — increase the amount of goods and services that the economy is capable of producing
    - increase in the productivity of these factors — more to be produced from the same amount of factors of production — result from improvements in technology or better organisation of production
- use of AD/AS diagrams to illustrate macroeconomic equilibrium
  - effect of a shift in aggregate demand on macroeconomic equilibrium
  - effect of a shift in short-run aggregate supply on macroeconomic equilibrium
  - Drawing an AD/AS diagram to illustrate the effect of an economic event
  - 和unit2描述图表差不多，但需要注意x轴(real national income/output),y轴(price level)
- effects of demand-side and supply-side shocks on the macroeconomy
  - effects of a demand-side shock
    - extent to which real national income and the price level are affected by the shock will depend on the current state of the economy

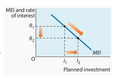
effects of a positive demand-side shock

  - effects of a supply-side shock
    - 去看causes that shifts in the SRAS curve

effects of a positive supply-side shock

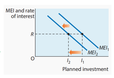
7.3 Determinants of aggregate demand

- meaning and components of aggregate demand
  - Aggregate demand is total planned expenditure on goods and services in the economy
  - AD=I+G+C+X-M
- determinants of consumption, investment, government spending, exports and imports
  - consumption is the largest single component of aggregate demand
  - determinants of consumption
    - income
      - MPC=ΔC/ΔY — marginal propensity to consume(MPC)
    - wealth
      - Rising house prices
      - increase in the price of shares
      - make it easier for households to borrow to purchase consumer durables — makes people more confident — increase in wealth
    - taxation
      - Disposable income = gross income + cash welfare benefits – taxes on income
    - consumer confidence
      - expectations about thee future
    - interest rates
      - interest rates increase — more rewarding for people to save — more expensive to borrow — reduce consumption — reduce household spending on consumer durables
    - the avaliability of credit
      - monetary policy can affect it — times when interest rates are low but some households may find it difficult to find a bank or other organisation that is willing and able to provide credit
  - determinants of investment
    - cost of the capital
      - invest to replace capital that has depreciated because it has worn out or is obsolete — Net investment = gross investment – depreciation
      - invest to increase the size of the capital stock and increase productive capacity
      - interest rates
        - rise in interest rates — increases the cost of borrowing to finance investment — even if the firms do not need to borrow money, opportunity cost of funds needs to be considered
        - marginal effciency of investment: the rate of return on an additional unit of investment — MEI>cost of funds — more profitable



effect of a fall in interest rates on planned investment

      - factors that shift MEI
        - cost of capital
        - change in confidence



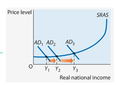
effect of a fall in the marginal efficiency of investment on planned investment

    - animal sirits — expectations and confidence
    - growth of national income
      - national income is growing — aggregate demand will increase — increase production to meet the increase in demand — more likely to invest in new capital to increase their productive capacity
    - amount of spare capacity
      - approach to full capacity — increase investment
    - changes in technology
      - improvements in techonoloy — improve efficiency — remain competitive
    - profits and the availability of finance — relunctant to lend fund — even a profitable project will not take places
    - taxes and subsidies
      - reduce taxes& increase subsidies — greater incentives to invest
  - determinants of exports
    - determine the volume and value of exports
      - inflation higher — relative prices — exports fall as they are less competitive
      - exchange rate
        - high inflation — fall in the exchange rate — compensate for the decline in competitiveness due to high inflation
      - quality and other non-price aspects of competitiveness — reputation and the standard of the service
      - national income and rate of growth in the major export markets
        - countries are growing rapidly and income are increasing — increase demand for foreign goods — increase exports
      - restrictions on international trade — tarrifs and quotas — decrease volume of exports
  - determinants of imports
    - 和determinants of exports一样的
  - determinants of government spending
    - social and political factors
    - the amount it can or is willing to raise in taxes
    - ability and willingness to borrow
    - wants to influence aggregate demand
- determinants of savings
  - income
    - saving ratio — household savings as a percentage of household disposable income
  - interest rates
    - consider real interest rater than nominal interest rate — the rate of interest after the effects of inflation have been removed
  - expectations and confidence
    - increase confidence — job and income will secure — saving fall
  - age
    - Older people often use their savings to support their consumption after they have stopped working
  - taxation
    - state provides unemployment benefits for unemployed people — less need to save
    - pays generous pensions for those in retirement
    - provision of services such as healthcare and education
  - government provision of welfare — developed financial system — trust in banks and other financial institutions
  - financial structure of the economy — ecourage people to save — but may also allow people to borrow more
  - social attitudes — in some societies saving is seen as being good and borrowing foolish
  - help finance investment spending and contribute to economic growth
- marginal propensities to consume, save, tax and imports
  - marginal propensity to save (MPS)
    - the proportion of a change in income that is saved — MPS=ΔS/ΔY — closed economy without a government sector — MPC+MPS=1
  - marginal propensity to tax
    - …that is taxed — MPT=ΔT/ΔY
  - marginal propensity to import
    - …that is spent on import — MPM=ΔM/ΔY — open economy with a government sector — MPC+MPS+MPT+MPM=1
- difference between saving and investment
  - Saving is a withdrawal from the circular flow of income
  - Investment is an injection into the circular flow of income and increases aggregate demand
- The basic accelerator process
  - I = α × ΔY
    - higher growth rate of output and/or a higher MCOR leads to a larger increase in investment — invest in capital goods to maintain the desired capital-output ratio
  - firms are working close to full capacity
    - will not be able to meet the increase in demand — unless invest to increase their productive capacity — How much they need to invest depends on the marginal capital-output ratio (MCOR)
      - the amount that has to be spent on capital to produce a given increase in annual output
      - MCOR=ΔK/ΔY
  - explains why changes in the level of national income affect investment
  - Limitations of the Accelerator Model
    - Assumes Fixed Capital-Output Ratio — adjust production techniques or operate with excess capacity — rather than immediately investing in new capital
    - Ignores Other Factors — Investment decisions are also influenced by other factors — interest rates, business confidence, and technological advancements — basic accelerator model does not consider
    - Time Lags — delays due to planning and implementation
    - does not consider replacement investment — replace existing capital as it wears out or becomes obsolete

7.4 Aggregate demand and the level of economic activity

- role of aggregate demand in influencing the level of economic activity
  - Real GDP is the main indicator of economic activity
  - In the short run, the main determinant of economic activity is AD
    - increase in aggregate demand — encourage firms to increase output to meet the increase in demand — empolyment increases
    - full capacity — firms find it difficult to increase production — price level is likely to rise — too much aggregate demand will cause inflation
- effect of an increase in aggregate demand on the level of economic activity

  - aggregate demand increases from AD1 to AD2 — real national income will rise from Y1 to Y2 — increase in aggregate demand only affects economic activity
  - if the economy is at Y3, there is very little spare capacity — real national income rises from Y3 to Y4, but the main effect is in inflation — increase in aggregate demand from AD3 to AD4 has little effect on economic activity
  - In the long run, production can only continue to increase if productive capacity is increasing — Increases in productive capacity result from supply-side improvements — aggregate demand can also affect economic activity in the long run — Unit 8相讲
  - Most governments attempt to manage aggregate demand to try to keep the economy producing close to its full capacity level of output
- The multiplier process
  - Multiplier=1/1-MPC
    - ΔY = ΔJ × multiplier
      - larger the MPC, the larger the size of the multiplier
        - In a closed economy without a government — Multiplier=1/MPS
        - in a open economy with a governmnt — Multiplier=1/(MPS+MPT+MPM)
  - withdrawals reduce the size of the multiplier because they reduce the amount spent on homeproduced goods and services
    - because they reduce the amount spent on homeproduced goods and services
  - fall in injections will also have a multiplier effect
  - The extent to which change in injections affect national income
    - 1. **Initial Injection**: The government spends $100 million (e.g., on infrastructure).
    - 2. **Aggregate Demand Rises**: This directly increases demand by $100 million.
    - 3. **Production Increases**: Firms produce more goods/services, raising national output by $100 million.
    - 4. **Income Generation**: Workers and businesses earn $100 million in new income.
    - 5. **Spending and Saving**: Households spend 67% (MPC = 0.67) of the new income ($67 million) and save 33%.
    - 6. **Repeat Cycles**: The $67 million becomes new demand, restarting the process. Each cycle adds progressively smaller amounts to national income.
    - 7. **Total Effect**: After all cycles, national income increases by $300 million (3x the initial injection).
    - multiplier effect arises because one person's spending becomes another's income
    - an injection of government spending initially increases aggregate demand from AD1 to AD2 — real national income from Y1 to Y2 — multiplier effect of the injection of government spending leads to a further increase in aggregatedemand, to AD3 — eventually increase real national income to Y3
    - if the economy is close to full capacity — not have much effect on real national income — increase the price level rather than national output



The multiplier effect

7.5 Determinants of short-run aggregate demand

- price level and production costs are the main determinants of short-run aggregate supply
  - depend on whether or not it is profitable
  - changes in costs will shift the SRAS curve
    - increase in cost of production — price level will have to rise to produce the same output
  - firms are producing close to full capacity — extra cost of increasing output rises — increase in the price level may be needed to make it profitable — movement on the SRAS curve

7.6 Determinants of short-run aggregate supply

- fundamental determinants of long-run aggregate supply
  - increase in the supply of factors of production and/or improvements in productivity will increase an economy's productive capacity
  - investment increases the size of capital stock and increase productivity
  - Attitudes to work and adaptability to change can affect the supply of labour and the productivity of the labour force
    - mobility of labour will affect the number of people who are unemployed as a result of changes
  - Economic incentives also affect the supply of labour
    - replacement ratio: the income received when unemployed as a ratio of the income received when in work — high replacement ratio — economic inventives to find work will be low
- vertical LRAS curve represents the normal capacity level of output of the economy
  - normal capacity level of output: the maximum output that an economy can continue to produce in the long run without experiencing inflationary pressures
  - if the economy benefits from long-run economic growth — normal capacity level of output will increase — LRAS curve shifts to right — increase supply of factor of production and/or productivity of factor of production — produce more goods and services without experiencing inflationary pressures
  - economy will not be able to continue to produce in equilibrium Ye in the long run — inflationary pressures — real national output/income will fall
    - there will be shortage of factors of production — increase wages — increase cost of production — increases prices and profits

An economy producing more than its normal capacity level of output in the short run

Presented with xmind By Dada Du